

DATA WAREHOUSE

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3.1 Introduction

The term "Data Warehouse" was first coined by Bill Inmon in 1990. According to Inmon, a data warehouse is a subject oriented, integrated, time-variant and non-volatile collection of data. This data helps analysts to take informed decisions in an organization. An operational database

undergoes frequent changes on a daily basis on account of the transactions that take place. Suppose a business executive wants to analyze previous feedback on any data such as a product, a supplier, or any consumer data, then the executive will have no data available to analyze because the previous data has been updated due to transactions.

A data warehouse provides us generalized and consolidated data in a multidimensional view. Along with generalized and consolidated view of data, a data warehouse also provides us Online Analytical Processing (OLAP) tools. These tools help us in interactive and effective analysis of data in a multidimensional space. This analysis results in data generalization and data mining. Data mining functions such as association, clustering, classification, prediction can be integrated with OLAP operations to enhance the interactive mining of knowledge at multiple levels of abstraction. That's why data warehouse has now become an important platform for data analysis and online analytical processing. A data warehouse is an architecture constructed by integrating data from multiple heterogeneous sources to support structured adhoc queries, analytical reporting and decision making. *A data warehouse is a subject-oriented, integrated, time-variant and nonvolatile collection of data in support of management's decision-making process.*

3.2 Understanding a Data Warehouse

A data warehouse is a database, which is kept separate from the organization's operational database. There is no frequent updating done in a data warehouse. A data warehouse possesses consolidated historical data, which helps the organization to analyze its business. A data warehouse helps executives to organize, understand, and use their data to take strategic decisions. Data warehouse systems help in the integration of diversity of application systems.

A data warehouse system helps in consolidated historical data analysis. A data warehouse does not require transaction processing, recovery and concurrency controls, because it is physically stored and separate from the operational database.

3.3 Data Warehouse Vs. Operational Database

- A data warehouse is kept separate from operational databases due to the following reasons. An operational database is constructed for well-known tasks and workloads such as searching particular records, indexing, etc. In contrast, data warehouse queries are often complex and they present a general form of data.
- Operational databases support concurrent processing of multiple transactions. Concurrency control and recovery mechanisms are required for operational databases to ensure robustness and consistency of the database.
- An operational database query allows to read and modify operations, while an OLAP query needs only read only access of stored data.
- An operational database maintains current data. On the other hand, a data warehouse maintains historical data.

3.4 Features of a Data Warehouse

The key features of a data warehouse are given below.

Subject Oriented – A data warehouse is subject oriented because it provides information around a subject rather than the organization's ongoing operations. These subjects can be product, customers, suppliers, sales, revenue, etc. A data warehouse does not focus on the ongoing operations, rather it focuses on modelling and analysis of data for decision making. Data warehouse provides a simple and concise view around particular subject issues by excluding data that are not useful in the decision support process.

- ✓ Perform simple transformations into a structure similar to the one in the data warehouse.

The data is extracted from the operational databases or the external information providers. Gateways are the application programs that are used to extract data. It is supported by underlying DBMS and allows a client program to generate SQL to be executed at a server. Open Database Connection (ODBC), Java Database Connection (JDBC), are examples of gateway.

Middle Tier: In the middle tier, we have the OLAP server that can be implemented in either of the following ways.

- ✓ By Relational OLAP (ROLAP), which is an extended relational database management system. The ROLAP maps the operations on multidimensional data to standard relational operations. Relational OLAP servers are placed between relational back-end server and client front-end tools. To store and manage the warehouse data, the relational OLAP uses relational or extended-relational DBMS. ROLAP includes the following.

- Implementation of aggregation navigation logic.
- Optimization for each DBMS back-end.
- Additional tools and services.

ROLAP servers are highly scalable. ROLAP tools analyze large volumes of data across multiple dimensions. ROLAP tools store and analyze highly volatile and changeable data.

- ✓ By Multidimensional OLAP (MOLAP) model, which directly implements the multi-dimensional data and operations. Multidimensional OLAP (MOLAP) uses array-based multidimensional storage engines for multidimensional views of data. With multi-

dimensional data stores, the storage utilization may be low if the dataset is sparse. Therefore, many MOLAP servers use two levels of data storage representation to handle dense and sparse datasets. MOLAP has the following characteristics.

- MOLAP tools process information with consistent response time regardless of level of summarizing or calculations selected.
- MOLAP tools need to avoid many of the complexities of creating a relational database to store data for analysis.
- MOLAP tools need fastest possible performance.
- MOLAP server adopts two level of storage representation to handle dense and sparse data sets.
- Denser sub-cubes are identified and stored as array structure.
- Sparse sub-cubes employ compression technology.

Top Tier: This tier is the front-end client layer. This layer holds the query tools and reporting tools, analysis tools and data mining tools. Fig. 2.11 depicts the three-tier architecture of data warehouse.

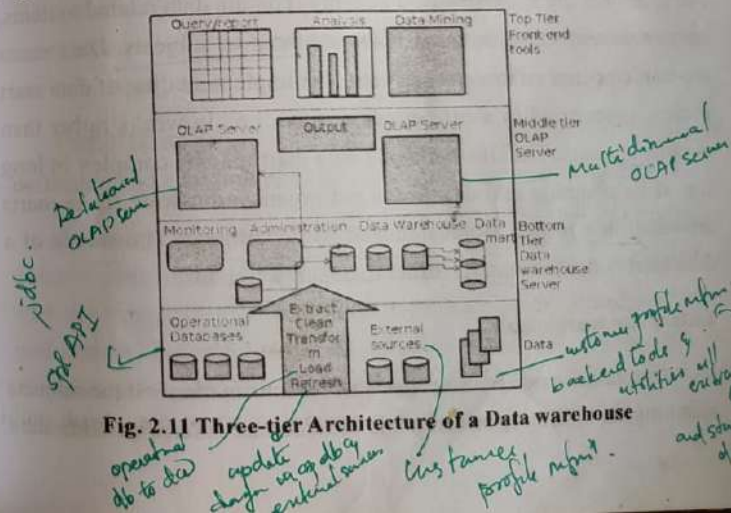


Fig. 2.11 Three-tier Architecture of a Data warehouse

The above 3-D table can be represented as 3-D data cube as shown in the following Fig. 2.1.

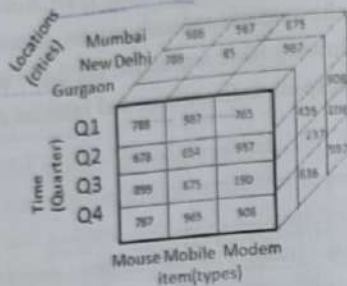


Fig. 2.1 3-D Data Cube

3.8 Conceptual Modelling of Data Warehouse

A data warehouse is based on a multidimensional data model which views data in the form of a data cube. A data cube, such as sales, allows data to be modeled and viewed in multiple dimensions. Dimension tables may be such as item (item_name, brand, type), or time (day, week, month, quarter, year). Fact table contains measures (such as dollars_sold) and keys to each of the related dimension tables. In data warehousing literature, an n-D base cube is called a base cuboid. The top most 0-D cuboid, which holds the highest-level of summarization, is called the apex cuboid. The lattice of cuboids is shown in Fig. 2.2 which forms a data cube.

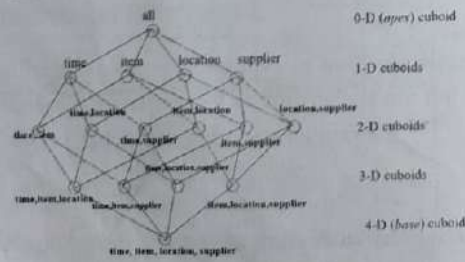


Fig. 2.2 Lattice of Cuboids

Schema is a logical description of the entire database. It includes the name and description of records of all record types including all associated data-items and aggregates. Much like a database, a data warehouse also requires to maintain a schema. A database uses relational model, while a data warehouse uses Star, Snowflake, and Fact Constellation schema. In this chapter, we will discuss the schemas used in a data warehouse.

Star Schema

Each dimension in a star schema is represented with only one-dimension table. This dimension table contains the set of attributes. Fig. 2.3 shows the star schema for sales data of a company with respect to the four dimensions, namely time, item, branch and location.

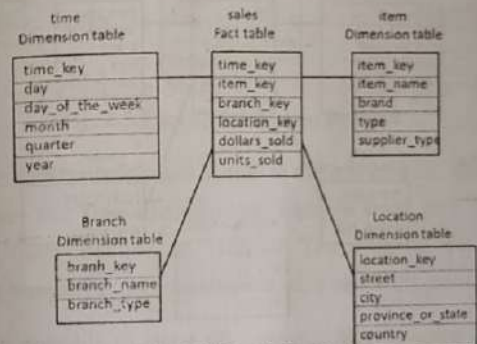


Fig. 2.3 Star Schema

There is a fact table at the center. It contains the keys to each of four dimensions. The fact table also contains the attributes, namely dollars_sold and units_sold. Each dimension has only one dimension table and each table holds a set of attributes. For example, the location dimension table contains the attribute set {location_key, street, city, province_or_state, country}. This constraint may cause data redundancy.

mechanisms, such as locking and logging, are required to ensure the consistency and robustness of transactions. An OLAP query often needs read-only access of data records for summarization and aggregation. Concurrency control and recovery mechanisms, if applied for such OLAP operations, may affect the execution of concurrent transactions and thus substantially reduce the throughput of an OLTP system.

Finally, the separation of operational databases from data warehouses is based on the different structures, contents and uses of the data in these two systems. Decision support requires historical data, whereas operational databases do not typically maintain historical data. In this context, the data in operational databases is usually far from complete for decision making. Decision support requires consolidation (such as aggregation and summarization) of data from heterogeneous sources, resulting in high-quality, clean and integrated data. In contrast, operational databases contain only detailed raw data, such as transactions, which need to be consolidated before analysis. Because the two systems provide quite different functionalities and require different kinds of data, it is presently necessary to maintain separate databases. However, many vendors of operational relational database management systems are beginning to optimize such systems to support OLAP queries. As this trend continues, the separation between OLTP and OLAP systems is expected to decrease.

3.7 Multidimensional Data Model

A data cube helps us represent data in multiple dimensions. It is defined by dimensions and facts. The dimensions are the entities with respect to which an enterprise preserves the records.

Illustration of Data Cube

Suppose a company wants to keep track of sales records with the help of sales data warehouse with respect to time, item, branch and

location. These dimensions allow to keep track of monthly sales and at which branch the items were sold. There is a table associated with each dimension. This table is known as dimension table. For example, "item" dimension table may have attributes such as item_name, item_type, and item_brand. The following table represents the 2-D view of Sales data for a company with respect to time, item and location dimensions.

Time(quarter)	Location="New Delhi"			
	Item(type)			
	Entertainment	Keyboard	Mobile	Locks
Q1	500	700	10	300
Q2	769	765	30	476
Q3	987	489	18	659
Q4	666	576	40	539

But here in this 2-D table, we have records with respect to time and item only. The sales for New Delhi are shown with respect to time, and item dimensions according to type of items sold. If we want to view the sales data with one more dimension, say, the location dimension, then the 3-D view would be useful. The 3-D view of the sales data with respect to time, item, and location is shown in the table below.

Time	Location="Gurgaon"			Location="New Delhi"			Location="Mumbai"		
	Item			Item			Item		
	Mouse	Mobile	Modem	Mouse	Mobile	Modem	Mouse	Mobile	Modem
Q1	788	987	765	786	85	987	986	567	875
Q2	678	654	987	659	786	436	500	876	948
Q3	809	875	150	983	909	232	947	100	1189
Q4	787	960	908	537	567	836	837	920	987

- Roll-up
- Drill-down
- Slice
- Dice
- Pivot (rotate)

Roll-up

Roll-up performs aggregation on a data cube in any of the following ways.

- ✓ By climbing up a concept hierarchy for a dimension.
- ✓ By dimension reduction.

Fig. 2.6 illustrates how roll-up works. Roll-up is performed by climbing up a concept hierarchy for the dimension location. Initially the concept hierarchy was "street < city < province < country". On rolling up, the data is aggregated by ascending the location hierarchy from the level of city to the level of country. The data is grouped into cities rather than countries. When roll-up is performed, one or more dimensions from the data cube is removed.

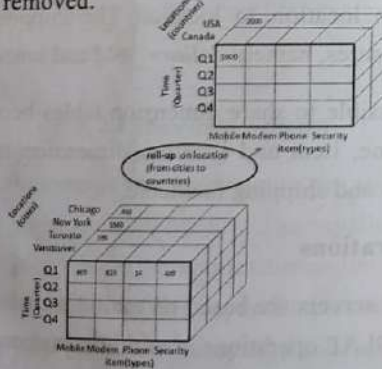


Fig. 2.6 Roll-up Operation

Drill-down

Drill-down is the reverse operation of roll-up. It is performed by either of the following ways.

- ✓ By stepping down a concept hierarchy for a dimension.
- ✓ By introducing a new dimension.

Fig. 2.7 illustrates how drill-down works. Drill-down is performed by stepping down a concept hierarchy for the dimension time. Initially the concept hierarchy was "day < month < quarter < year". On drilling down, the time dimension is descended from the level of quarter to the level of month. When drill-down is performed, one or more dimensions from the data cube are added. It navigates the data from less detailed data to highly detailed data.

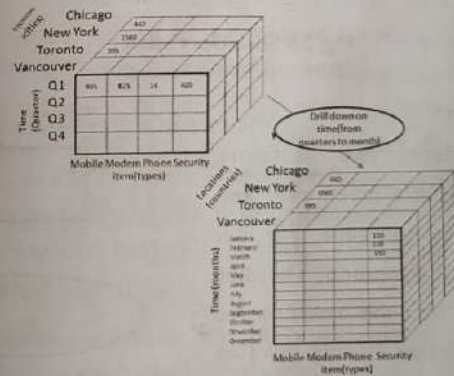


Fig. 2.7 Drill-Down Operation

Slice

The slice operation selects one particular dimension from a given cube and provides a new sub-cube. Fig. 2.8 shows how slice operation

Operational metadata: It includes currency of data and data lineage. *transferred to logs* Currency of data means whether the data is active, archived, or purged. *removed* Lineage of data means the history of data migrated and transformation applied on it.

3.13 Indexing OLAP Data

The bitmap indexing method is popular in OLAP products because it allows quick searching in data cubes. The bitmap index is an alternative representation of the record ID (RID) list. In the bitmap index for a given attribute, there is a distinct bit vector B_v for each value v in the domain of the attribute. If the domain of a given attribute consists of n values, then n bits are needed for each entry in the bitmap index (i.e., there are n bit vectors). If the attribute has the value v for a given row in the data table, then the bit representing that value is set to 1 in the corresponding row of the bitmap index. All other bits for that row are set to 0. But this is not suitable for high cardinality domains. Fig. 2.13 shows an example for bitmap indexing.

Base table			Index on Region			Index on Type			
Cust	Region	Type	RecID	Asia	Europe	America	RecID	Retail	Dealer
C1	Asia	Retail	1	1	0	0	1	1	0
C2	Europe	Dealer	2	0	1	0	2	0	1
C3	Asia	Dealer	3	1	0	0	3	0	1
C4	America	Retail	4	0	0	1	4	1	0
C5	Europe	Dealer	5	0	1	0	5	0	1

Fig. 2.13 Bitmap Indexing

In the above figure, the base table contains 3 fields namely cust_id, region and type. In order to speed up the searching process, bitmap indices are created on region and type.

3.14 Review Questions

1. What is a data warehouse?
2. What are the features of a data warehouse?
3. Compare and contrast OLAP and OLTP.
4. Explain multidimensional data cube.
5. Explain the various OLAP operations.
6. Explain the conceptual modelling of a data warehouse.
7. Explain the three-tier architecture of a data warehouse.
8. What is a data mart?
9. What are the data warehouse models?
10. Explain bit-map indexing on OLAP data.
11. Explain the importance of metadata repository.
12. Explain ROLAP and MOLAP servers.

works. Here slice is performed for the dimension "time" using the criterion time = "Q1". It will form a new sub-cube by selecting one or more dimensions.

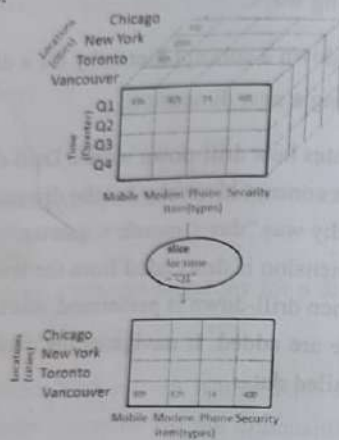


Fig. 2.8 Slice Operation

Dice

Dice selects two or more dimensions from a given cube and provides a new sub-cube. Fig. 2.9 shows the dice operation. The dice operation on the cube based on the following selection criteria involves three dimensions.

(location = "Toronto" or "Vancouver")

(time = "Q1" or "Q2")

(item = "Mobile" or "Modem")

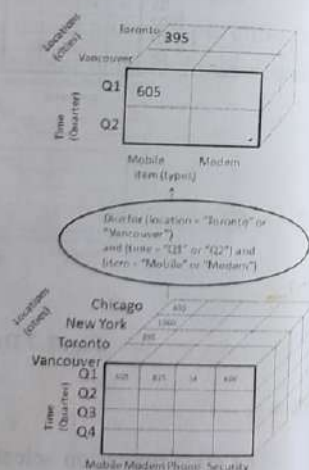


Fig. 2.9 Dice Operation

Pivot

The pivot operation is also known as rotation. It rotates the data axes in view in order to provide an alternative presentation of data. Fig. 2.10 shows the pivot operation.

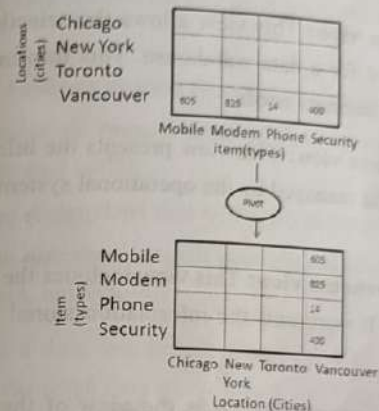


Fig. 2.10 Pivot Operation

3.10 Business Analysis Framework

The business analyst get the information from the data warehouses to measure the performance and make critical adjustments in order to win over other business holders in the market. Having a data warehouse offers the following advantages.

- ✓ Since a data warehouse can gather information quickly and efficiently, it can enhance business productivity.
- ✓ A data warehouse provides us a consistent view of customers and items, hence, it helps us manage customer relationship.
- ✓ A data warehouse also helps in bringing down the costs by tracking trends, patterns over a long period in a consistent and reliable manner.

For example, "Vancouver" and "Victoria" are cities in the Canadian province of British Columbia. The entries for such cities may cause data redundancy along the attributes province_or_state and country.

Snowflake Schema

Some dimension tables in the snowflake schema are normalized. The normalization splits up the data into additional tables. Unlike star schema, the dimensions table in a snowflake schema are normalized. For example, the item dimension table in star schema is normalized and split into two dimension tables, namely item and supplier table. Snowflake schema is shown in Fig. 2.4.

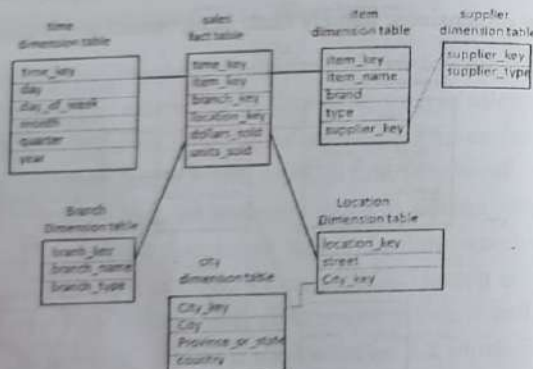


Fig. 2.4 Snowflake Schema

Now the item dimension table contains the attributes item_key, item_name, type, brand and supplier-key. The supplier key is linked to the supplier dimension table. The supplier dimension table contains the attributes supplier_key and supplier_type. Due to normalization in the Snowflake schema, the redundancy is reduced and therefore, it becomes easy to maintain and save storage space.

Fact Constellation Schema

A fact constellation has multiple fact tables. It is also known as galaxy schema. Fig. 2.5 shows fact constellation schema. It shows two fact tables, namely sales and shipping.

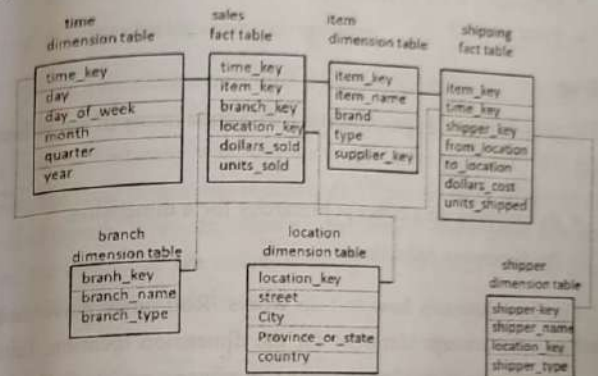


Fig. 2.5 Fact Constellation Schema

The sales fact table is same as that in the star schema. The shipping fact table has the five dimensions, namely item_key, time_key, shipper_key, from_location, to_location. The shipping fact table also contains two measures, namely dollars_sold and units_sold.

It is also possible to share dimension tables between fact tables. For example, time, item and location dimension tables are shared between the sales and shipping fact table.

3.9 OLAP Operations

Since OLAP servers are based on multidimensional view of data, we will discuss OLAP operations in multidimensional data. Following is the list of OLAP operations.

To design an effective and efficient data warehouse, we need to understand and analyze the business needs and construct a business analysis framework. Each person has different views regarding the design of a data warehouse. These views are as follows.

✓ **The top-down view:** This view allows the selection of relevant information needed for a data warehouse. This information matches current and future business needs.

✓ **The data source view:** This view presents the information being captured, stored and managed by the operational system. It is modeled by E-R Model.

✓ **The data warehouse view:** This view includes the fact tables and dimension tables. It represents the information stored inside the data warehouse.

✓ **The business query view:** It is the view of the data from the viewpoint of the end-user.

Building a data warehouse is a complex task because it requires **business skills**, **technology skills** and **program management skills**. **Business skills** include addressing the following questions.

- ✓ How systems store and manage their data?
- ✓ How to build extractors that transfer data from the operational system to the data warehouse?
- ✓ How to build warehouse refresh software?

Technology skills address how data analysts make assessments from quantitative information and derive conclusions. These skills include ability to discover patterns and trends, to extrapolate trends based on history and look for anomalies. **Program management skills** are needed

to interface with many technologies, vendors and end users in order to deliver results in a timely and cost effective manner.

3.11 Three-Tier Architecture

A typical data warehouse design process involves the following four steps.

- ✓ Choose a **business process** to model, e.g., orders, invoices, etc.
- ✓ Choose the **grain (atomic level of data)** of the business process (data in the fact table).
- ✓ Choose the **dimensions** that will apply to each fact table record.
- ✓ Choose the **measure** that will populate each fact table record (e.g. dollars_sold, units_sold).

Generally a data warehouse adopts a three-tier architecture. Following are the three tiers of the data warehouse architecture.

Bottom Tier: The bottom tier of the architecture is the data warehouse server. It is the relational database system. We use the back end tools like load manager and utilities to feed data into the bottom tier. These back end tools and utilities perform the extract, clean, load, and refresh functions.

Load Manager

This component performs the operations required to extract and load process. The size and complexity of the load manager varies between specific solutions from one data warehouse to other. The load manager performs the following functions.

- ✓ Extract the data from source system.
- ✓ Load the extracted data into temporary data store very fast.

Integrated – A data warehouse is constructed by integrating data from heterogeneous sources such as relational databases, flat files, etc. This integration enhances the effective analysis of data. Data cleaning and data integration techniques are applied and during this process, it ensures consistency in naming conventions, encoding structures, attribute measures, etc. among different data sources. For e.g., Hotel price: currency, tax, breakfast covered, etc.

Time Variant – The data collected in a data warehouse is identified with a particular time period. The data in a data warehouse provides information from the historical point of view. The time horizon for the data warehouse is significantly longer than that of operational systems. Operational database deals with current value data. Data in a data warehouse provide information from a historical perspective (e.g., past 5-10 years). Every key structure in the data warehouse contains an element of time, explicitly or implicitly. But the key of operational database may or may not contain "time element".

Non-volatile – Non-volatile means the previous data is not erased when new data is added to it. A data warehouse is kept separate from the operational database and therefore frequent changes in operational database is not reflected in the data warehouse. A data warehouse requires only two operations in data accessing: initial loading of data and access of data.

3.5 OLAP Vs. OLTP

OLAP stands for On Line Analytical Processing which takes place in a data warehouse. OLTP stands for On Line Transaction Processing which takes place in an operational data base. Following table shows comparison of OLAP and OLTP.

Data Mining

	OLTP	OLAP
Feature	operational processing transaction	informational processing analysis
Characteristic	clerk, DBA, database professional	knowledge worker (e.g., manager, executive, analyst)
Orientation	day-to-day operations	long-term informational requirements, decision support
Function	PR based, application-oriented	star/snowflake, subject-oriented
DB design	current, guaranteed up-to-date	historical, accuracy maintained over time
Data	primitive, highly detailed	summarized, consolidated
Summarization	detailed, flat relational	summarized, multidimensional
View	short, simple transaction	complex query
Unit of work	read/write	mostly read
Access	data in	information out
Focus	index/hash on primary key	lots of scans
Operations		
Number of records accessed	tens	millions
Number of users	thousands	hundreds
DB size	100 MB to GB	100 GB to TB
Priority	high performance, high availability	high flexibility, end-user autonomy
Metric	transaction throughput	query throughput, response time

3.6 Need for a Separate Data Warehouse

Operational databases store huge amounts of data. But on-line analytical processing is not performed directly on such databases. A major reason for such a separation is to help promote the high performance of both systems. An operational database is designed and tuned from known tasks and workloads, such as indexing and hashing using primary keys, searching for particular records and optimizing queries. On the other hand, data warehouse queries are often complex. They involve the computation of large groups of data at summarized levels and may require the use of special data organization, access and implementation methods based on multidimensional views. Processing OLAP queries in operational databases would substantially degrade the performance of operational tasks.

Moreover, an operational database supports the concurrent processing of multiple transactions. Concurrency control and recovery

Data Warehouse Models

From the perspective of data warehouse architecture, we have the following data warehouse models.

- Virtual warehouse
- Data mart
- Enterprise warehouse

Virtual warehouse

The view over an operational data warehouse is known as a virtual warehouse. It is easy to build a virtual warehouse. Building a virtual warehouse requires excess capacity on operational database servers.

Data mart

Data mart contains a subset of organization-wide data. This subset of data is valuable to specific groups of an organization. In other words, we can claim that data marts contain data specific to a particular group. For example, the marketing data mart may contain data related to items, customers and sales. Data marts are confined to subjects. Data marts are implemented on low-cost servers. The implementation of data mart cycles is measured in short periods of time, i.e., in weeks rather than months or years. The life cycle of a data mart may be complex in long run, if its planning and design are not organization-wide. Data marts are small in size and are customized by department. The source of a data mart is departmentally structured data warehouse.

Enterprise warehouse

An enterprise warehouse collects all the information and the subjects spanning an entire organization. It provides us enterprise-wide data

The data is integrated from operational systems and external information providers. This information can vary from a few gigabytes to hundreds of gigabytes, terabytes or beyond.

3.12 Meta Data Repository

Metadata is simply defined as data about data. The data that is used to represent other data is known as metadata. For example, the index of a book serves as a metadata for the contents in the book. In other words, we can say that metadata is the summarized data that leads us to detailed data. In terms of data warehouse, we can define metadata as follows.

- ✓ Metadata is the road-map to a data warehouse.
- ✓ Metadata in a data warehouse defines the warehouse objects.
- ✓ Metadata acts as a directory. This directory helps the decision support system to locate the contents of a data warehouse.

In a data warehouse, we create metadata for the data names and definitions of a given data warehouse. Along with this metadata, additional metadata is also created for time-stamping any extracted data and the source of extracted data. Metadata can be broadly categorized into three categories.

Business metadata: It has the data ownership information, business definition and charging policies.

Technical metadata: It includes database system names, table and column names, sizes, data types and allowed values. Technical metadata includes structural information such as primary and foreign key attributes and indices. Technical metadata also contains the algorithms used for summarization, the mapping from operational environment to the data warehouse and data related to system performance.